

PROJECT PROPOSAL
ENHANCEMENT OF BATTERY-POWERED RICKSHAWS

BY
TEAM MURUBBEEE

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Overview:

We, team Murubbeee, aim to work on enhancing battery-run rickshaws in Bangladesh. Battery rickshaws have become a common urban transportation in Bangladesh, especially in busy cities like Dhaka, and Chittagong. According to Dhaka Tribune's report on May 24, 2024, almost 60 lakh battery rickshaws are operating on the streets of Bangladesh. These rickshaws provide us with the benefits of fast transportation as compared to their counterpart, human-powered rickshaws. However, these benefits come with a price. Battery rickshaws are the major reasons for accidents, excessive electricity usage, and traffic-related issues. To address these issues and also upgrade these battery-run rickshaws for more efficient results, we propose our newly modified battery rickshaws.

Problem Statement:

Battery-run rickshaws are the cause of a good number of problems in the urban areas of our country. The major issues that we will be addressing are:

1. Unnecessary speeding of battery rickshaws.
2. Collisions that lead to minor or severe injuries and life threats
3. Excessive use of electricity for charging the battery-run rickshaws
4. Lack of safety measurements in the rickshaws
5. Absence of an information system in the rickshaw.

Battery-run rickshaws are most criticized for their poor safety measurements and functionality. The major issue is that these rickshaws do not have any safety options to control overspeeding. The speeds these rickshaws can gain are often too hard to control with the brakes they are equipped with. These lead to collisions with other vehicles on the road and serious injuries, sometimes even death. The inclusion of speed sensors and automatic response systems for overspeeding can resolve this issue to an extent.

Another problem that the battery-run rickshaws create is the electricity they consume for charging. A study conducted in 2023 shows that the battery-run rickshaws consume up to 450 MW of electricity every day. However, this number might be even larger as there are many illegal connections used for charging these rickshaws. To address this issue, the use of renewable energy systems and the addition of a hybrid mechanism is proposed.

Outline of the proposed solution:

To solve the mentioned problem statements above, we aim to use several sensors and a hybrid mechanism in battery-run rickshaws accompanied by the use of renewable sources to cater to power saving. The proposed modifications are:

1. Solar panel installation on the top of rickshaw hoods
2. Speed sensors to detect the speed of the rickshaw
3. Obstacle detector to be installed at the front portion
4. A system that can respond to overspeed and reduce chances of collisions by controlling the power supply to motor.
5. A reverse charging facility that can use the rotation of wheels to charge the battery
6. Addition of indicator lamps and headlamps
7. Information system (LED display) to show necessary data (speed, charge, etc.)

The prototype we aim to make will comprise of the following modules:

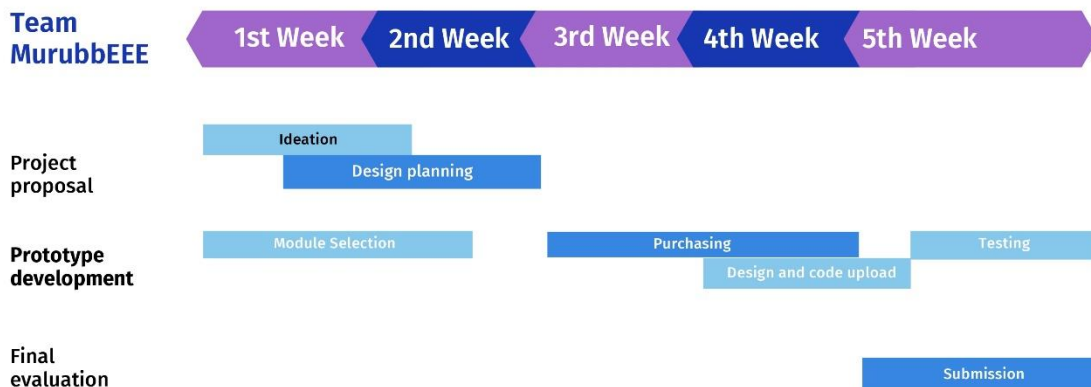
- Arduino
- Solar panel
- Speed measuring module
- Ultrasonic SONAR sensor (HC-SR04)
- Dual Motor drivers
- LCD module
- LED
- An alternator





Project timeline:

PROJECT ROADMAP



References:

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2. ENERGY CONSUMPTION BY BATTERY OPERATED AUTO-RICKSHAWS. (2013). *DAFFODIL INTERNATIONAL UNIVERSITY JOURNAL OF SCIENCE AND TECHNOLOGY*, 8(2), 71. <https://www.bip.org.bd/admin/uploads/member-publication/2013/202001250931511.pdf>
3. Issue-I, T. A. (2022, February 2). *Impact of ban on battery-run rickshaws in districts*. The Financial Express. <https://thefinancialexpress.com.bd/views/columns/impact-of-ban-on-battery-run-rickshaws-in-districts-1643812426>